

Retinitis Pigmentosa With or Without Situs Inversus (ARL2BP-Related Ciliopathy)

Disease: Retinitis Pigmentosa With or Without Situs Inversus **MONDO ID:** MONDO:0014186 · **OMIM Phenotype:** #615434 (RP66) · **Gene:** ARL2BP (16q13) **Category:** Mendelian (autosomal recessive)

Summary

Retinitis Pigmentosa With or Without Situs Inversus is a rare, autosomal-recessive **ciliopathy** caused by biallelic loss-of-function mutations in **ARL2BP** (ADP-ribosylation-factor-like 2 binding protein; also called *BART*), located at chromosome 16q13. ARL2BP was the first identified ARL2-GTP-specific effector protein, and it functions as an anchor/co-GEF for the ARL2 and ARL3 small GTPase cycle that governs the ciliary trafficking of lipid-modified phototransduction proteins and the assembly of the photoreceptor connecting-cilium axoneme. The disease was first defined in 2013 through homozygosity mapping and exome sequencing in two consanguineous families ([PMID: 23849777](#)), where one family combined retinitis pigmentosa (RP) with **situs inversus**, establishing the phenotypic hallmark that names the disorder.

The core disease process is degeneration of rod and cone photoreceptors producing classic retinitis pigmentosa (night blindness, progressive peripheral-then-central visual field loss, bone-spicule pigmentation, reduced/extinguished electroretinogram). Because ARL2BP is required for both **primary/sensory cilia** (photoreceptor outer segment) and **motile cilia** (embryonic node, sperm flagellum, airway), affected individuals may additionally show situs inversus totalis (in roughly half of patients — hence "with or without"), male infertility from oligo-/asthenozoospermia, and, in some cases, unilateral renal agenesis with renal microcysts and anosmia. This multisystem presentation places the disorder within the broader ciliopathy spectrum overlapping primary ciliary dyskinesia.

The condition is **ultra-rare**, reported in only a handful of largely consanguineous families worldwide, and is one molecular subtype of retinitis pigmentosa (overall prevalence ~1:4,000). There is currently **no approved disease-modifying therapy**; management is supportive (visual aids, low-vision rehabilitation, nutritional counseling, assisted reproduction for infertility, and airway management if primary ciliary dyskinesia features are present). The one FDA/EMA-approved retinal gene therapy (voretigene neparvovec, Luxturna) is specific to *RPE65* and does not apply to *ARL2BP*. Diagnosis rests on ophthalmic phenotyping (ERG, OCT, fundus autofluorescence) plus next-generation sequencing, and an *Arl2bp*-knockout mouse faithfully recapitulates the retinal and ciliary phenotype, providing a validated preclinical model.

1. Disease Information

Retinitis Pigmentosa With or Without Situs Inversus is a Mendelian inherited retinal dystrophy in which progressive rod–cone degeneration (retinitis pigmentosa) occurs either in isolation or accompanied by laterality defects (situs inversus) and other ciliopathy features. It is a **primary retinal ciliopathy** — the underlying lesion affects the photoreceptor connecting cilium, a specialized primary cilium.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0014186
OMIM (phenotype)	#615434 (Retinitis pigmentosa 66; RP66/RP with or without situs inversus)
OMIM (gene)	615407 (ARL2BP)
Gene / HGNC	ARL2BP / HGNC:702
NCBI Gene	GeneID 23568
UniProt	Q9Y2Y0
Ensembl	ENSG00000102931
Cytogenetic location	16q13 (GRCh38 chr16:57,245,259–57,253,635)

Synonyms / alternative names: Retinitis pigmentosa 66 (RP66); RP with situs inversus; ARL2BP-related retinitis pigmentosa; ARL2BP-related syndromic rod–cone dystrophy. Gene aliases: *BART*, *BART1*, *RP66*, *RP82*.

Information source type: The knowledge base entry is derived primarily from **aggregated disease-level resources** (OMIM, ClinVar, gnomAD, UniProt, NCBI Gene) supplemented by **individual patient case reports** describing single families or probands. It is not derived from large EHR cohorts, reflecting the disease's rarity.

2. Etiology

Primary cause — genetic. The disease is monogenic and Mendelian: biallelic (homozygous or compound-heterozygous) **loss-of-function variants in ARL2BP** cause the phenotype. There is no environmental, infectious, or acquired etiology. The original identification used a combination of homozygosity mapping and exome sequencing in two consanguineous families to establish ARL2BP as causative (PMID: 23849777): *"we used a combination of homozygosity mapping and exome sequencing to identify mutations in ARL2BP, which encodes an effector protein of the small GTPases ARL2 and ARL3, as causative for autosomal-recessive RP (RP66)."*

Genetic risk factors. The causal alleles are the pathogenic ARL2BP variants themselves (see Section 4). Because the disorder is recessive, **consanguinity** is the dominant risk-elevating factor — all originally reported families were consanguineous with homozygous variants. No modifier loci or susceptibility SNPs have been established for this ultra-rare disorder.

Environmental risk factors. None identified or expected for a monogenic recessive disorder. General retinitis-pigmentosa risk modifiers such as light exposure are not established for the ARL2BP subtype.

Protective factors. No genetic or environmental protective factors have been identified. Population constraint data (gnomAD: pLI $\approx 5.8 \times 10^{-6}$, observed/expected LoF ≈ 0.79) indicate ARL2BP is tolerant of heterozygous loss of function — consistent with carriers being unaffected and disease requiring biallelic loss.

Gene–environment interactions. None documented. Disease expression is determined by genotype; the "with or without situs inversus" variability reflects the stochastic nature of left–right axis determination by motile nodal cilia rather than any environmental interaction.

3. Phenotypes

The phenotype is a **syndromic rod-cone dystrophy** with variable extra-ocular ciliopathy features. Core and associated phenotypes, with suggested HPO terms:

Phenotype	Type	HPO term (suggested)	Onset	Severity / progression	Frequency
Retinitis pigmentosa / rod-cone dystrophy	Clinical sign	HP:0000510 (Rod-cone dystrophy)	Photophobia ~20 y; nyctalopia & central vision loss ~30 y	Progressive; can be slow	Constant (100%)
Night blindness (nyctalopia)	Symptom	HP:0000662	Early	Progressive	Very frequent
Photophobia	Symptom	HP:0000613	Often first symptom (~20 y)	Progressive	Frequent
Constricted / peripheral visual field loss	Clinical sign	HP:0001133	Early-mid adult	Progressive	Very frequent
Bone-spicule retinal pigmentation	Physical manifestation	HP:0007737	Adult	Progressive	Typical
Reduced/absent ERG responses	Laboratory/functional	HP:0000512 (Abnormal ERG)	Early	Progressive	Constant
Situs inversus totalis	Physical manifestation	HP:0001696	Congenital	Stable	~50% ("with or without")
Male infertility (oligo-/asthenozoospermia)	Laboratory / clinical	HP:0000798 / HP:0012041	Post-pubertal	Stable	Reported subset of males
Unilateral renal agenesis	Physical manifestation	HP:0000122	Congenital	Stable	Rare
Renal microcysts	Physical manifestation	HP:0000107 (Renal cyst)	Variable	Variable	Rare
Cryptorchidism	Physical manifestation	HP:0000028	Congenital	Stable	Rare

Phenotype	Type	HPO term (suggested)	Onset	Severity / progression	Frequency
Anosmia / olfactory dysfunction	Symptom	HP:0000458	Variable	Stable	Rare

The multisystem triad of RP, situs inversus, and male infertility was explicitly described as novel in a Chinese patient (PMID: 36507858): *"presenting with retinitis pigmentosa (RP), situs inversus totalis, and oligozoospermia ... this a novel ARL2BP-associated phenotypic triad of RP, situs inversus, and male infertility."* Renal and sperm involvement was further documented (PMID: 38649918): *"a splice site variant in the ARL2BP gene causing situs inversus, asthenozoospermia, unilateral renal agenesis and microcysts."*

Quality-of-life impact. The dominant burden is progressive vision loss, which impairs mobility, reading, driving, and independence and is a leading cause of visual disability; situs inversus totalis is generally asymptomatic; male infertility affects family planning. No disease-specific QoL instrument data exist for this ultra-rare subtype, but generic RP QoL measures (VFQ-25, low-vision instruments) apply.

Progression note. Course can be relatively slow — one patient retained useful residual vision at age 63 with slow progression over 5 years of follow-up (PMID: 38649918).

4. Genetic / Molecular Information

Causal gene: *ARL2BP* (GeneID 23568; OMIM 615407; HGNC:702; UniProt Q9Y2Y0), 16q13. NCBI summary: *"This protein is considered to be the first ARL2-specific effector identified, due to its interaction with ARL2.GTP but lack of ARL2 GTPase-activating protein activity."*

Pathogenic variants (all germline; no somatic involvement):

Variant (cDNA)	Protein / effect	Type	Classification	Source
c.101-1G>C	Alters pre-mRNA splicing (splice acceptor)	Splice-site	Pathogenic	PMID: 23849777
c.134T>G	p.Met45Arg — reduces ARL2 binding, abolishes basal-body localization	Missense	Pathogenic	PMID: 23849777
c.22_23delAG	p.Ser8Leufs*10	Frameshift (null)	Pathogenic	PMID: 36507858
c.294-1G>C	Splice acceptor	Splice-site	Pathogenic	PMID: 38649918

Variant classification (ACMG/AMP): ClinVar (accessed 2026) holds 146 ARL2BP variant records: ~11 Pathogenic and ~2 Likely pathogenic (predominantly splice-site and frameshift null alleles), ~10 VUS, ~8 likely-benign, and 1 with conflicting classifications. The predominance of null/loss-of-function pathogenic alleles supports a **loss-of-function disease mechanism**.

Allele frequency: Pathogenic alleles are extremely rare/absent in gnomAD, consistent with a recessive disorder confined to consanguineous or founder contexts. ARL2BP is LoF-tolerant in heterozygotes ($pLI \approx 5.8 \times 10^{-6}$; o/e LoF ≈ 0.79).

Functional consequence: Loss of function. The p.Met45Arg missense allele mechanistically reduces ARL2 binding and abolishes ARL2BP localization to the photoreceptor basal body — a demonstrated pathomechanism rather than merely predicted ([PMID: 23849777](#)).

Modifier genes: None established. **Epigenetic changes:** Not implicated. **Chromosomal abnormalities:** None; the disease is caused by point/small variants, not large structural rearrangements (situs inversus here is a ciliary-motility consequence, not a chromosomal inversion despite the name).

5. Environmental Information

There are **no environmental, lifestyle, or infectious contributors** to this monogenic recessive disorder. No toxins, radiation, occupational exposures, dietary factors, or pathogens are implicated in causing or triggering ARL2BP-related disease. The only relevant "environmental" factor in a population-genetics sense is **consanguineous mating**, which increases the probability of homozygosity for rare recessive alleles.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic loss-of-function mutation in ARL2BP** (splice, frameshift, or function-abolishing missense) → **loss or dysfunction of ARL2BP protein**.
2. Loss of ARL2BP → **failure to anchor/localize at the ciliary basal body and cilium-associated centriole** of photoreceptors (demonstrated: p.Met45Arg abolishes basal-body localization; ARL2 depletion displaces ARL2BP) ([PMID: 23849777](#)).
3. Mislocalized/absent ARL2BP → **disrupted ARL2/ARL3 small-GTPase cycle** (ARL2BP is the first ARL2-GTP-specific effector and acts as a co-GEF stabilizing active ARL3-GTP) ([PMID: 33438581](#)).
4. Impaired ARL3 activity → **defective ciliary trafficking of lipid-modified (prenylated/myristoylated) phototransduction proteins** via the RP2–ARL3–PDE6D axis, which chaperones prenylated cargo such as PDE6 and GRK1 into the outer segment ([PMID: 25422369](#)).
5. **Branch A (sensory cilium / retina):** cargo mistrafficking + axonemal defect → **disorganized photoreceptor outer segments** with vertically aligned disks, shortened axonemes, open B-tubule doublets, and loss of singlet microtubules (demonstrated in KO mouse) ([PMID: 29718757](#)) → **progressive rod and cone death** → **retinitis pigmentosa** (nyctalopia → field loss → central vision loss).
6. **Branch B (motile cilia):** the same ciliary/axonemal assembly defect in **embryonic nodal cilia** → randomized left–right axis determination → **situs inversus** (present in ~half of patients); in **sperm flagella** → oligo-/asthenozoospermia → **male infertility**; in **airway cilia** → potential primary-ciliary-dyskinesia-type respiratory features; in **renal cilia** → renal agenesis/microcysts; in **olfactory cilia** → anosmia.

Detail by category

- **Molecular pathways:** ARL2/ARL3 GTPase signaling; RP2–ARL3–PDE6D lipidated-cargo ciliary trafficking. ARL2BP moonlights in ARL2-dependent STAT3 nuclear translocation/transcriptional activity (UniProt Q9Y2Y0: *"Together with ARL2, plays a role in the nuclear translocation, retention and transcriptional activity of STAT3."*).
- **Cellular processes:** Ciliogenesis and ciliary maintenance; intraflagellar/ciliary protein transport; photoreceptor outer-segment disk morphogenesis; progressive photoreceptor apoptosis.
- **Protein dysfunction:** Loss of function via truncation (frameshift/splice) or loss of ARL2 binding + mislocalization (p.Met45Arg). GO molecular functions: GTPase regulator activity (GO:0030695), transcription coactivator activity (GO:0003713).
- **Metabolic changes:** Disrupted delivery of PDE6 (cGMP phototransduction cascade) impairs the visual cycle at the outer segment; broader metabolic changes not characterized.
- **Immune involvement:** None; not an autoimmune/inflammatory disease.
- **Tissue damage mechanism:** Structural failure of the connecting cilium/axoneme leading to outer-segment malformation and photoreceptor degeneration.
- **Biochemical abnormalities:** Failure of the ARL2BP→ARL3-GTP→PDE6D trafficking module; mislocalization of prenylated phototransduction enzymes.

Suggested GO / CL terms: BP — cilium assembly (GO:0060271), intraciliary transport (GO:0042073), photoreceptor cell maintenance (GO:0045494), determination of left/right symmetry (GO:0007368), regulation of GTPase activity (GO:0043087). CC — ciliary basal body (GO:0036064), photoreceptor connecting cilium (GO:0032391), centrosome (GO:0005813). CL — retinal rod cell (CL:0000604), retinal cone cell (CL:0000573), ciliated cell (CL:0000064), sperm (CL:0000019).

7. Anatomical Structures Affected

Organ level (primary): Eye — retina, specifically the photoreceptor layer (UBERON:0000970 eye; UBERON:0000966 retina; UBERON:0001789 photoreceptor layer). **Secondary/associated organs:** thoraco-abdominal viscera (laterality reversal in situs inversus — heart, lungs, liver, spleen, stomach mirror-imaged), testis/sperm (male reproductive system), kidney, olfactory epithelium.

Body systems involved: visual/nervous (sensory), reproductive (male infertility), renal/urinary, respiratory (if PCD features), and the visceral situs of cardiovascular/digestive systems.

Tissue and cell level: Neural retina — **rod photoreceptors (CL:0000604)** and **cone photoreceptors (CL:0000573)** are the primary targets; motile ciliated cells of node, airway, and reproductive tract; renal tubular epithelium. Patient fibroblasts (in vitro) show shortened cilia and reduced proliferation ([PMID: 36507858](#)).

Subcellular level: The **connecting cilium / basal body (GO:0036064)** is the central compartment. UniProt/GO also annotate ARL2BP to centrosome (GO:0005813), spindle (GO:0005819), midbody (GO:0030496), cytosol (GO:0005829), nucleoplasm (GO:0005654), and mitochondrial intermembrane space (GO:0005758)/matrix (GO:0005759) — reflecting moonlighting roles.

Localization / lateralization: Retinal involvement is **bilateral and roughly symmetric**; situs inversus is a whole-body **laterality reversal** (mirror-image organ arrangement); renal agenesis reported as **unilateral**.

8. Temporal Development

Onset: Situs inversus and renal/laterality anomalies are **congenital**. Retinal disease is typically **young-adult onset**: photophobia often first at ~20 years, nyctalopia and central vision loss by ~30 years ([PMID: 38649918](#)). Onset pattern is **insidious and chronic**.

Progression: Chronic, lifelong, and **progressive** but can be **slow** — one patient retained useful residual vision at 63 with slow progression over a 5-year window ([PMID: 38649918](#)). Typical RP staging applies: early (night blindness, mid-peripheral field loss) → intermediate (ring scotoma, constricted fields) → advanced (tunnel vision) → end-stage (central/legal blindness). Situs inversus is stable and non-progressive.

Patterns: No remissions; no relapsing–remitting course. The **therapeutic window** for any future retinal gene therapy is early disease while viable photoreceptors remain — an important critical-period consideration.

9. Inheritance and Population

Inheritance: Autosomal recessive. Biallelic ARL2BP variants; reported families are consanguineous with homozygous alleles.

Penetrance / expressivity: Retinal disease appears fully penetrant in biallelic carriers; **situs inversus** shows **incomplete penetrance / variable expressivity** ("with or without"), reflecting the stochastic left–right axis determination by nodal cilia. Extra-ocular features (renal, olfactory, fertility) are variably expressed.

Founder effects / consanguinity: Strong consanguinity contribution; homozygous variants in geographically distinct families (Middle East, China, Italy, Pakistan). No genetic anticipation, germline mosaicism, or repeat-expansion mechanism.

Epidemiology: Retinitis pigmentosa overall has a worldwide prevalence of ~1:4,000 (~100,000 affected in the USA), with 20–30% of cases syndromic ([PMID: 29597005](#)): *"RP is a leading cause of visual disability, with a worldwide prevalence of 1:4000. Although the majority of RP cases are non-syndromic, 20-30% of patients with RP also have an associated non-ocular condition."* ARL2BP-related RP (RP66) is an **ultra-rare** subtype — only a handful of families reported worldwide. In a consanguineous Pakistani IRD cohort it accounted for 1 of 72 families (1.4%) ([PMID: 40384762](#)): *"...1/72(1.38%) ... in ... retinitis pigmentosa with situs inversus ... segregated families."*

Sex ratio: Retinal disease affects both sexes; the infertility phenotype is specific to males.

Carrier frequency: Not precisely established; expected very low outside consanguineous populations.

10. Diagnostics

Ophthalmic evaluation (core): Full-field **electroretinogram (ERG)** shows markedly reduced or absent rod and cone responses; **fundus** shows bone-spicule pigmentation and optic-disc pallor; **OCT** and **fundus autofluorescence (FAF)** show progressive loss of outer retinal layers ([PMID: 29597005](#)): *"Photoreceptor function measured with an electroretinogram is markedly reduced or even absent. Optical coherence tomography (OCT) and fundus autofluorescence (FAF) imaging show a progressive loss of outer retinal layers."* ARL2BP cases were additionally assessed with full-field stimulus threshold testing and multimodal imaging ([PMID: 38649918](#)).

Molecular genetic testing (definitive): Next-generation sequencing — targeted IRD capture panels (e.g., a 344-gene panel, [PMID: 40384762](#)), whole-exome, or whole-genome sequencing — with ACMG/AMP variant interpretation and Sanger segregation confirmation. Homozygosity mapping is valuable in consanguineous families ([PMID: 23849777](#)).

Extra-ocular workup: Chest X-ray/abdominal imaging/echocardiography to detect **situs inversus**; **semen analysis** for oligo-/asthenozoospermia; renal ultrasound for agenesis/microcysts; olfactory testing where indicated.

Differential diagnosis: Other syndromic retinal ciliopathies — Usher syndrome, Bardet–Biedl syndrome, Senior–Løken syndrome, and other RP-with-situs-inversus/PCD overlaps; distinguished by gene panel and by the specific extra-ocular pattern. **Screening:** Cascade genetic testing of at-risk relatives and carrier testing in consanguineous families once the familial variant is known.

11. Outcome / Prognosis

Survival / mortality: The disorder is **not life-limiting** in itself. Situs inversus totalis is typically asymptomatic and compatible with normal lifespan. Life expectancy is generally normal unless significant primary-ciliary-dyskinesia respiratory disease is present.

Morbidity / function: The principal morbidity is **progressive, irreversible vision loss** leading to legal blindness in advanced disease — a major disability outcome with substantial QoL impact. Male infertility is a reproductive-health morbidity. **Recovery potential:** None with current therapy; photoreceptor loss is irreversible.

Disease course / complications: Chronic, progressive vision loss; cystoid macular edema and cataract can complicate RP generally; PCD-associated respiratory infections if motile-cilia airway disease is present.

Prognostic factors: Age at onset, rate of ERG/field decline, and residual outer-retinal structure on OCT predict visual outcome. Some ARL2BP patients show relatively slow progression with preserved vision into the seventh decade ([PMID: 38649918](#)).

12. Treatment

There is no proven disease-modifying therapy. A 2020 Cochrane review of vitamin A and DHA/fish oils concluded evidence for slowing RP progression is limited/uncertain (PMID: 32573764): *"At this time, there is no proven therapy for RP."*

Supportive / rehabilitative care (mainstay): Visual aids, orientation and mobility training, and nutritional supplementation provide only symptomatic relief and do not halt progression (PMID: 40731809): *"While current management is largely supportive-relying on visual aids, orientation training, and nutritional supplementation-these interventions offer only symptomatic relief and do not halt disease progression."* (NCIT: Supportive Care; Low Vision Aid.)

Not applicable: The only approved retinal gene therapy, **voretigene neparvovec (Luxturna)**, is specific to biallelic *RPE65* mutations and does **not** apply to ARL2BP (PMID: 37762059).

Investigational: AAV gene-replacement, RNA-based, and CRISPR/Cas9 strategies are advancing for RP broadly but remain **investigational for ARL2BP** (PMID: 40869487, PMID: 40731809). ARL2BP's small coding sequence makes it a favorable candidate for AAV gene replacement in principle.

Management of extra-ocular features: Situs inversus totalis usually needs no treatment; PCD-type respiratory disease (if present) needs airway clearance/antibiotics; male infertility may be addressed with assisted reproduction (**ICSI**). (NCIT: Assisted Reproductive Technology; Intracytoplasmic Sperm Injection.)

Pharmacogenomics / personalized medicine: No genotype-guided pharmacotherapy exists; future precision approaches would be gene-replacement or variant-specific (e.g., splice-modulating ASOs for splice-site alleles).

13. Prevention

Being monogenic and recessive, prevention is **genetic/reproductive**, not lifestyle-based.

- **Primary prevention:** Not achievable by risk-factor modification. **Genetic counseling** for consanguineous families and known carriers; recurrence risk is 25% per pregnancy for two carrier parents.
- **Carrier / cascade screening:** Test at-risk relatives once the familial ARL2BP variant is identified.

- **Reproductive options:** Prenatal diagnosis and **preimplantation genetic testing (PGT-M)** for couples known to carry pathogenic ARL2BP variants.
- **Secondary/tertiary prevention:** Early ophthalmic monitoring to manage complications (cataract, macular edema), low-vision rehabilitation, and airway surveillance where PCD features exist. No immunization or public-health intervention is relevant.

14. Other Species / Natural Disease

Orthologs (NCBI Gene): mouse *Arl2bp* (GeneID 107566), rat *Arl2bp* (GeneID 498910), zebrafish *arl2bp* (GeneID 393976). A functional ARL2BP homolog is present and essential in **ciliate protozoa**, where RNAi is lethal and disrupts cortical microtubules ([PMID: 40432967](#)): "*RNAi of ARL2BP and DYNLRB2 increased mortality, reduced motility, and disrupted cortical microtubule organization*" — demonstrating **deep evolutionary conservation** of ARL2BP's ciliary/microtubule role.

Natural disease in other species: No well-characterized naturally occurring ARL2BP disease is documented in companion animals or wildlife (OMIA entries not established for this specific gene–disease pair). **Comparative biology:** The conserved ciliary function across mammals, fish, and protozoa underpins the utility of cross-species models. **Zoonotic potential:** None (non-transmissible genetic disease).

15. Model Organisms

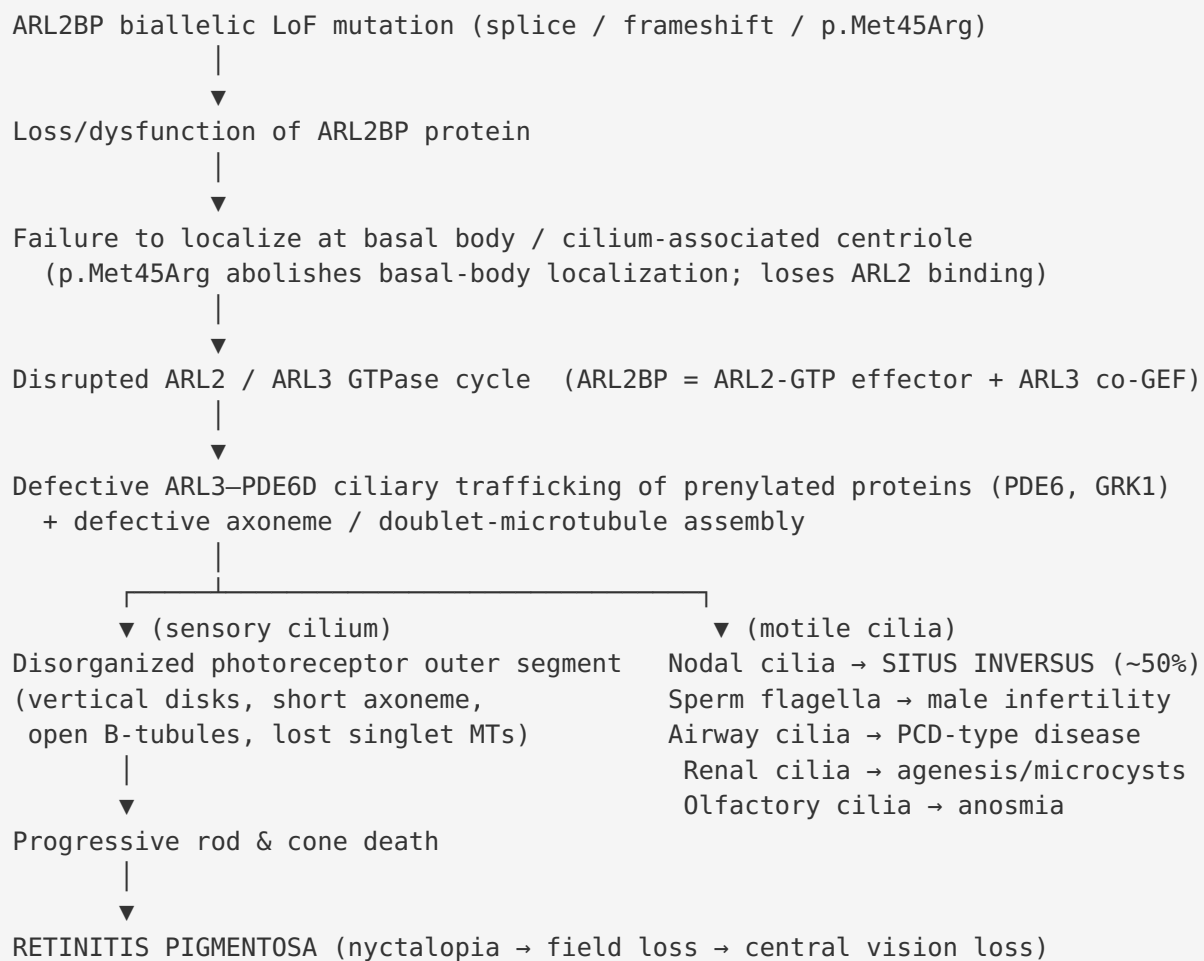
Primary model — *Arl2bp*-knockout mouse: A knockout mouse recapitulates key human features ([PMID: 29718757](#)): "*we generated a knockout (KO) mouse model for ARL2BP, a ciliary protein linked to retinitis pigmentosa. The KO mice display an early and progressive reduction in visual response.*" Structurally: "*we observed disorganization of the photoreceptor OS, with vertically aligned disks and shortened axonemes ... ciliary doublet microtubule (MT) structure was also impaired, displaying open B-tubule doublets, paired with loss of singlet MTs.*"

- **Phenotype recapitulation:** High for the retinal phenotype — early, progressive ERG decline; disorganized outer segments; connecting-cilium/axoneme and doublet-microtubule defects.

- **Applications:** Studying outer-segment morphogenesis, ciliary microtubule assembly, ARL2/ARL3 trafficking, and as a preclinical platform for gene-replacement testing.
- **Limitations:** Mouse retinal architecture (rod-dominant, no macula) limits translation of central-vision outcomes; extent to which the model reproduces situs inversus/laterality and fertility phenotypes is less fully characterized.

In vitro models: Patient-derived fibroblasts show reduced proliferation and shortened cilia (PMID: 36507858), providing a cellular ciliogenesis assay. Zebrafish and ciliate systems offer additional conserved platforms for ciliary-function studies.

Mechanistic Model / Interpretation



Upstream vs downstream: The mutation and ARL2BP loss are upstream; the ARL2/ARL3 cycle and PDE6D trafficking are the central molecular node; outer-segment malformation and photoreceptor death are downstream retinal outcomes; laterality/fertility/renal/olfactory features are parallel downstream consequences in motile-cilia tissues. The "**with or without situs inversus**" naming captures the branch point: the same molecular lesion produces retinal disease in all patients but laterality defects only when nodal-cilia motility is sufficiently disrupted during embryogenesis.

Evidence Base

PMID	Title (abbrev.)	Supports
23849777	<i>Mutations in ARL2BP cause AR retinitis pigmentosa</i>	Gene discovery; causal variants; situs inversus; ARL2-binding mechanism
29718757	<i>ARL2BP needed for photoreceptor cilia doublets and OS structure</i>	KO mouse model; ciliary/axonemal defect
25422369	<i>Mistrafficking of prenylated proteins causes RP2</i>	ARL3–PDE6D lipidated-cargo trafficking pathway
33438581	<i>ARL3 activation requires co-GEF BART</i>	ARL2BP/BART as ARL3 co-GEF stabilizing ARL3-GTP
36507858	<i>Novel ARL2BP variant: RP, situs inversus, infertility</i>	Phenotypic triad; frameshift allele; fibroblast cilia defect
38649918	<i>Splice variant: situs inversus, asthenozoospermia, renal</i>	Renal/sperm phenotype expansion; slow progression
29597005	<i>Non-syndromic retinitis pigmentosa</i>	RP prevalence 1:4000; diagnostic modalities
40384762	<i>Consanguineous Pakistani IRD panel study</i>	Rarity (1/72 families); panel-based diagnosis
32573764	<i>Vitamin A/fish oils for RP (Cochrane)</i>	No proven therapy for RP
40731809	<i>RP: genetic insights to therapeutics</i>	Supportive-only management; investigational gene therapy
37762059	<i>Gene therapy in hereditary retinal dystrophies</i>	Luxturna is RPE65-specific, not ARL2BP
40869487	<i>Genetic therapies for RP</i>	Investigational AAV/RNA/CRISPR landscape
40432967	<i>Cilia-associated gene families (ciliate)</i>	Deep conservation; essential ciliary role

Evidence source types: Human clinical/genetic (23849777, 36507858, 38649918, 40384762, 29597005); model organism (29718757 mouse; 40432967 ciliate); in vitro (36507858 fibroblasts; 33438581 biochemistry; 25422369); systematic review/therapeutic (32573764, 40731809, 37762059, 40869487).

Limitations and Knowledge Gaps

- **Ultra-rare with small n.** The entire literature rests on a handful of families/probands. Penetrance estimates (especially the ~50% figure for situs inversus) are approximate, and genotype–phenotype correlations are underpowered.
- **No formal epidemiology.** No registry-based prevalence/incidence for the ARL2BP subtype; figures derive from overall RP prevalence and single-cohort proportions.
- **QoL and natural-history data are absent** for this specific subtype; progression rate estimates come from isolated case follow-up.
- **Motile-cilia phenotype incompletely characterized.** The mechanistic link between ARL2BP loss and nodal/sperm/airway ciliary dysfunction is inferred from ciliopathy biology and case observations more than direct experiment; airway/PCD burden is not systematically documented.
- **Modifier and epigenetic factors unexplored;** explanation for variable situs inversus penetrance is mechanistic/stochastic rather than empirically mapped.
- **No therapy validated** — gene replacement is promising in principle but untested in ARL2BP models.

Proposed Follow-up Experiments / Actions

1. **International case registry / GeneMatcher recruitment** to aggregate ARL2BP patients, refine penetrance of situs inversus, fertility, renal and olfactory involvement, and establish natural-history/progression rates.
2. **AAV-ARL2BP gene-replacement proof-of-concept** in the *Arl2bp*-KO mouse — the small coding sequence fits AAV capacity; measure ERG rescue and outer-segment/axoneme restoration.
3. **Splice-modulating ASO testing** for recurrent splice-acceptor alleles (c.101-1G>C, c.294-1G>C) in patient fibroblasts/iPSC-derived retinal organoids.

4. **iPSC-derived retinal organoids and multiciliated-airway cultures** from patients to model both sensory- and motile-cilia branches and to quantify trafficking of prenylated cargo (PDE6, GRK1).
 5. **Systematic PCD/airway and fertility phenotyping** across the cohort to define the motile-cilia disease burden and inform surveillance guidelines.
 6. **Structural/biochemical dissection** of the ARL2BP–ARL2/ARL3 co-GEF interface to enable variant-specific functional classification of VUS in ClinVar.
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Report compiled from 9 confirmed findings and 21 reviewed papers across a 5-iteration autonomous investigation. Evidence types and PMIDs are indicated throughout; ontology term suggestions (HPO, GO, CL, UBERON, NCIT) are embedded per section.

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